Probing the Limits: Adversarial and Out-of-Distribution Robustness of Neural Thicket Ensembles in the Vision Domain

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Abstract

Neural thickets — dense clusters of task-solving expert solutions reachable via random weight perturbations around pretrained models — offer a remarkably cheap approach to ensemble construction. While this phenomenon has been demonstrated for large language models on reasoning tasks, its implications for adversarial robustness and out-of-distribution (OOD) detection remain unexplored. We transplant the neural thicket ensemble concept to the vision domain, constructing ensembles of up to 10 members by perturbing a pretrained ResNet-18 on CIFAR-10. We evaluate three perturbation strategies (isotropic Gaussian, orthogonal, and layer-scaled) under white-box PGD-20 and FGSM attacks, and assess OOD detection performance on SVHN using energy scores and mutual information. Our results reveal a fundamental limitation: vision models exhibit a sharp sensitivity cliff, where perturbation scales large enough to create meaningful diversity ($\sigma \geq 0.02$) catastrophically destroy accuracy (94% $\rightarrow$ 28%), while scales that preserve accuracy ($\sigma \leq 0.001$) produce near-identical ensemble members with <1% disagreement. Consequently, thicket ensembles achieve PGD-20 accuracy of 12.5%, indistinguishable from a single model (12.6%), while adversarial training reaches 78.5%. Neither orthogonal nor layer-scaled perturbations overcome this limitation. We conclude that the neural thicket phenomenon may be specific to heavily overparameterized models, and that weight perturbation alone is insufficient for adversarial robustness in vision.

1 Introduction

Can we build robust ensembles for free? The recent discovery of *neural thickets* [Gan and Isola, 2026] suggests a tantalizing possibility: pretrained model weights are surrounded by dense clusters of task-specific expert solutions, reachable via simple random Gaussian perturbations with zero gradient-based training. If these ensembles also resist adversarial attacks and detect out-of-distribution inputs, they would offer an extraordinarily cheap path to reliable AI systems.

Ensemble methods have long been recognized as a practical defense against adversarial examples [Pang et al., 2019] and a natural mechanism for uncertainty quantification [Fort et al., 2019]. The intuition is straightforward: diverse models are unlikely to share the same failure modes, so majority voting or prediction averaging can filter out adversarial perturbations that fool individual members. However, conventional deep ensembles require training multiple models independently — an expensive proposition that scales linearly with ensemble size.

Neural thickets [Gan and Isola, 2026] offer a radical alternative. The RANDOPT algorithm generates N random weight perturbations $\boldsymbol{\theta}' = \boldsymbol{\theta} + \sigma\cdot$ around pretrained weights $\boldsymbol{\theta}$, evaluates each on a validation set, selects the top-K by accuracy, and ensembles their predictions. On LLM reasoning benchmarks, this approach matches reinforcement learning methods at a fraction of the cost. The key insight is that large pretrained models sit within a "thicket" of nearby solutions — a region of weight space dense with functional alternatives.

But are these alternatives truly diverse? For thicket ensembles to provide robustness, members must disagree on adversarial or out-of-distribution inputs even when they agree on clean data. This requires functional diversity — not merely proximity in weight space. Prior work on ensemble diversity for robustness [Pang et al., 2019, Zhao et al., 2024] shows that diversity must be explicitly encouraged, as independently trained models often converge to similar decision boundaries. The situation may be even more challenging for thicket ensembles, where members share the same pretrained initialization and differ only by small random perturbations.

We investigate this question by transplanting neural thicket ensembles to the vision domain (RESNET-18 on CIFAR-10), where adversarial robustness is well-studied and standard attack protocols exist. We test three perturbation strategies — isotropic Gaussian, orthogonal, and layer-scaled — across five perturbation scales, and evaluate adversarial robustness under PGD-20 and FGSM attacks as well as OOD detection on SVHN.

Our results reveal a fundamental negative finding: vision models have much tighter loss basins than LLMs, creating a sharp phase transition where perturbation scales large enough for diversity ($\sigma \geq 0.02$) catastrophically destroy model accuracy (94% $\rightarrow$ 28%). At viable scales ($\sigma \leq 0.001$), ensemble members are near-clones with $<$1% disagreement, providing no robustness benefit. Thicket ensembles achieve PGD-20 accuracy of 12.5% — indistinguishable from a single model — while adversarial training reaches 78.5%.

We make the following contributions:

- We provide the first evaluation of neural thicket ensembles for adversarial robustness and OOD detection, testing three perturbation strategies across five scales.
- We identify a sharp sensitivity cliff in vision models that fundamentally limits weight-perturbation ensembles: no regime exists where both high accuracy and high diversity coexist.
- We demonstrate that adversarial training (78.5% PGD accuracy) remains vastly superior to any ensemble strategy (12–20%), and that even deep ensembles with true mode diversity offer only modest robustness gains.
- We provide evidence that the neural thicket phenomenon may be scale-dependent, with implications for when weight-perturbation ensembles can and cannot be expected to work.

2 Related Work

Neural thickets and weight-space ensembles. Gan and Isola [2026] discover that pretrained LLM weights are surrounded by dense clusters of task-specific expert solutions, reachable via random Gaussian perturbations. Their RANDOPT algorithm selects top-K perturbed models by validation accuracy and ensembles predictions via majority vote, achieving performance competitive with PPO and GRPO on reasoning benchmarks at $O(1)$ training cost. The same research group previously explored task-aware ensemble construction through SPLIT-ENSEMBLE [Chen et al., 2024], which splits classification into complementary subtasks for OOD-aware training. Our work extends neural thickets to the vision domain and provides the first evaluation of their adversarial robustness and OOD detection capabilities.

Model soups and weight interpolation. A related line of work explores linear combinations of model weights for robustness. Croce et al. [2023] show that "model soups" — convex or extrapolated combinations of adversarially fine-tuned models — can smoothly trade off robustness to different ℓ_p-norm adversaries without retraining. Ramé et al. [2022] demonstrate that averaging weights of diverse models improves OOD generalization. These approaches operate in the same weight-space paradigm as neural thickets, but use structured directions (fine-tuned models) rather than random perturbations. Our work reveals why the random perturbation approach fails for vision: the perturbation must be orders of magnitude smaller than the structured directions used in model soups.

Ensemble diversity for adversarial robustness. The connection between ensemble diversity and adversarial robustness has been extensively studied. Pang et al. [2019] propose Adaptive Diversity Promoting (ADP) regularization, which encourages non-maximal predictions to be mutually orthogonal across ensemble members, reducing adversarial transferability. Zhao et al. [2024] integrate low-curvature models for ensemble defense, showing that diversity combined with flat loss landscapes improves robustness. Hu et al. [2024] propose median-based weight averaging for self-

ensemble adversarial training. A common finding across these works is that diversity must be explicitly encouraged during training — incidental diversity from different random seeds or perturbations is typically insufficient [Fort et al., 2019]. Our experiments confirm this finding in the extreme case of weight-perturbation ensembles.

OOD detection with ensembles. Deep ensembles are a natural tool for OOD detection, as ensemble disagreement can signal distributional shift. However, Xia and Bouganis [2022] demonstrate that ensemble diversity (measured by mutual information) is not reliably useful for OOD detection at scale; instead, averaging task-specific scores (e.g., energy) over ensemble members is more effective. Our OOD detection results corroborate this finding: energy-based detection consistently outperforms mutual information across all thicket variants.

Loss landscape geometry and robustness. Fort et al. [2019] show that deep ensembles derive their uncertainty estimates from exploring different modes of the loss landscape, and that the quality of uncertainty quantification depends on mode diversity rather than proximity in weight space. Lu et al. [2024] identify a tradeoff between sharpness of individual learners and ensemble diversity. These geometric perspectives help explain our negative result: perturbations within a single sharp basin of the loss landscape cannot access the diverse modes needed for robustness.

3 Methodology

We adapt the RANDOPT framework [Gan and Isola, 2026] from LLMs to the vision domain, constructing neural thicket ensembles around a pretrained RESNET-18 on CIFAR-10. We then evaluate these ensembles for adversarial robustness and OOD detection.

3.1 Problem Setup

Let $f_{\boldsymbol{\theta}} : \mathcal{X} \to \mathbb{R}^C$ denote a classifier with parameters $\boldsymbol{\theta} \in \mathbb{R}^d$ mapping inputs $\boldsymbol{x} \in \mathcal{X}$ to logits over C classes. Given a pretrained model $f_{\boldsymbol{\theta}^*}$, we construct an ensemble $\{f_{\boldsymbol{\theta}'_1}, \ldots, f_{\boldsymbol{\theta}'_K}\}$ by perturbing $\boldsymbol{\theta}^*$ and selecting the top-K models by validation accuracy:

$$\boldsymbol{\theta}'_i = \boldsymbol{\theta}^* + \sigma \cdot {}_i, \quad {}_i \sim p(), \tag{1}$$

where $\sigma > 0$ controls the perturbation scale and $p()$ is a perturbation distribution. From a pool of $N = 60$ candidates, we select the K models with highest validation accuracy and ensemble their predictions via average softmax probabilities.

3.2 Perturbation Strategies

We test three perturbation distributions $p()$:

Isotropic Gaussian. The baseline strategy from Gan and Isola [2026]: $\sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$. Each parameter receives an independent perturbation drawn from a standard normal distribution, scaled by σ.

Orthogonal perturbations. We enforce approximate orthogonality among perturbation directions via Gram-Schmidt orthogonalization. This ensures that ensemble members are maximally spread in weight space, potentially increasing functional diversity.

Layer-scaled perturbations. We scale the perturbation magnitude per layer proportional to the layer's weight norm:

$$\boldsymbol{\theta}'_l = \boldsymbol{\theta}_l + \sigma \cdot \frac{\|\boldsymbol{\theta}_l\|}{\sqrt{d_l}} \cdot {}_l, \quad {}_l \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}), \tag{2}$$

where $\boldsymbol{\theta}_l$ denotes the parameters of layer l and d_l is their dimensionality. This normalizes the relative perturbation magnitude across layers with different weight scales.

3.3 Base Model Training

We train a RESNET-18 adapted for CIFAR-10 (3×3 initial convolution, no max pooling) with approximately 11.2M parameters. Training uses SGD with learning rate 0.1, momentum 0.9, weight decay 5×10^{-4}, and cosine annealing over 50 epochs. Data augmentation includes random cropping (32×32, padding 4) and random horizontal flips. We use a 45K/5K train/validation split, with the standard 10K test set held out for evaluation. The base model achieves 94.1% test accuracy.

Table 1: Adversarial robustness comparison. Thicket ensembles ($\sigma = 0.001$, $K = 5$) provide no improvement over a single model, while adversarial training (PGD-AT) achieves 78.5% PGD accuracy. Best results in **bold**.

Method	Clean Acc (%)	PGD-20 Acc (%)	FGSM Acc (%)
Single Model	94.1	12.6	47.8
Gaussian THICKET	94.0	12.5	47.4
Orthogonal THICKET	94.0	12.4	47.2
Layer-Scaled THICKET	94.1	12.6	47.8
DEEP ENSEMBLE (3 members)	**94.7**	19.7	53.2
PGD-AT	91.2	**78.5**	**79.7**

3.4 Adversarial Evaluation

We evaluate robustness under two white-box attacks, both with ℓ_∞ threat model:

PGD-20. Projected Gradient Descent [Madry et al., 2018] with $\epsilon = 8/255$, step size $\alpha = 2/255$, and 20 iterations. For ensembles, the attack optimizes against the average softmax output.

FGSM. Fast Gradient Sign Method as a single-step baseline with $\epsilon = 8/255$.

3.5 OOD Detection

We evaluate OOD detection using CIFAR-10 as the in-distribution dataset and SVHN as the OOD dataset, both normalized with CIFAR-10 statistics. We compute three uncertainty scores:

Maximum Softmax Probability (MSP). $S_{\text{MSP}}(\boldsymbol{x}) = \max_c \bar{p}_c(\boldsymbol{x})$, where $\bar{p}(\boldsymbol{x})$ is the average softmax over ensemble members.

Energy score. $S_{\text{E}}(\boldsymbol{x}) = -\log \sum_c \exp(\bar{z}_c(\boldsymbol{x}))$, where $\bar{z}(\boldsymbol{x})$ is the average logit vector.

Mutual Information (MI). $\text{MI}(\boldsymbol{x}) = H[\bar{p}(\boldsymbol{x})] - \frac{1}{K}\sum_{k=1}^{K} H[p_k(\boldsymbol{x})]$, measuring ensemble disagreement as the gap between total and average per-member entropy.

We report AUROC and FPR@95 (false positive rate at 95% true positive rate).

3.6 Baselines

We compare against three baselines:

- **Single model**: The pretrained RESNET-18 with no perturbation.
- **Deep ensemble**: 3 independently trained RESNET-18 models with different random seeds, representing true mode diversity.
- **Adversarial training (PGD-AT)**: A RESNET-18 trained with 7-step PGD adversarial training for 30 epochs — the gold standard for adversarial robustness.

4 Results

4.1 Adversarial Robustness

Table 1 presents the main adversarial robustness results. All three thicket variants achieve PGD-20 accuracy within 0.2% of the single model, providing no meaningful robustness improvement. At $\sigma = 0.001$ — the largest viable perturbation scale — ensemble members have a disagreement rate of only 0.6%, meaning they are functionally identical. The white-box PGD attacker trivially finds adversarial examples that fool all members simultaneously.

The deep ensemble achieves 19.7% PGD accuracy — a 56% relative improvement over the single model — demonstrating that genuine functional diversity from independent training trajectories does confer some robustness. However, this is still far from the 78.5% achieved by adversarial training, confirming that diversity alone is insufficient for strong adversarial robustness.

Table 2: Perturbation scale σ controls a sharp tradeoff between accuracy and diversity. A phase transition between $\sigma = 0.01$ and $\sigma = 0.02$ leaves no regime where both high accuracy and high diversity coexist.

σ	Clean Acc (%)	Mean Candidate Acc (%)	Disagreement Rate
0.001	94.0	94.1	0.006
0.005	93.8	93.6	0.033
0.01	92.4	89.9	0.091
0.02	28.4	14.0	0.614
0.05	11.7	10.2	0.830

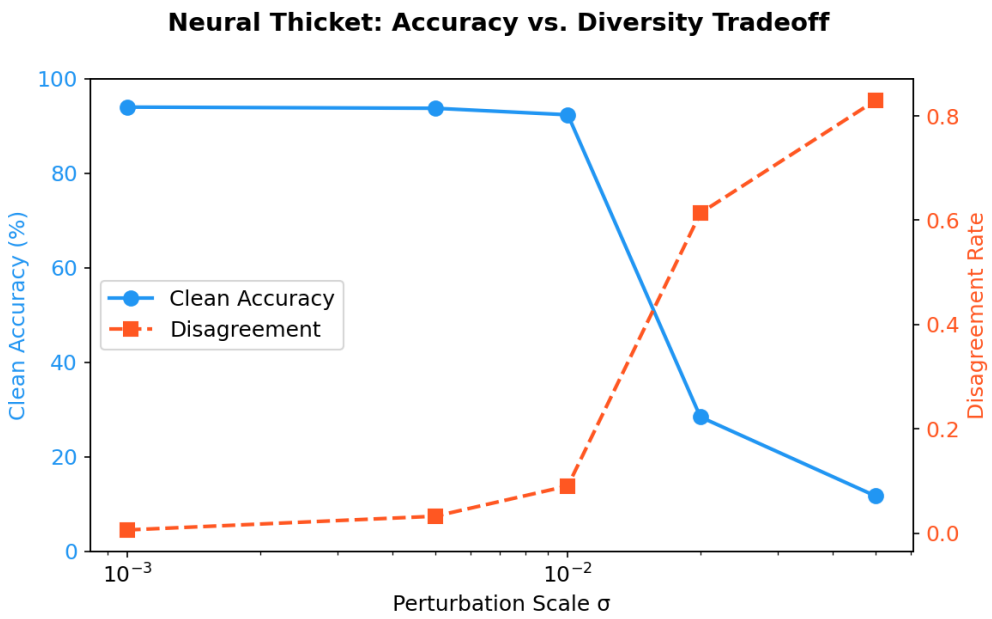


Figure 1: Accuracy vs. diversity across perturbation scales σ. A sharp phase transition occurs between $\sigma = 0.01$ and $\sigma = 0.02$: accuracy collapses from 92.4% to 28.4% while disagreement jumps from 9.1% to 61.4%. No useful intermediate regime exists.

4.2 The Sensitivity Cliff

Table 2 reveals the fundamental constraint underlying our negative result. There is a sharp phase transition between $\sigma = 0.01$ (92.4% accuracy, 9.1% disagreement) and $\sigma = 0.02$ (28.4% accuracy, 61.4% disagreement). The model is catastrophically destroyed by perturbations of magnitude $\sim$0.02, which corresponds to roughly 2% of parameter magnitudes. No intermediate regime exists where both high accuracy and meaningful diversity coexist.

Figure 1 visualizes this tradeoff, showing the cliff-like behavior rather than the gradual tradeoff one might expect. This contrasts sharply with LLMs, where Gan and Isola [2026] report that models with billions of parameters tolerate substantially larger perturbations.

4.3 OOD Detection

Table 3 presents OOD detection results for CIFAR-10 (in-distribution) vs. SVHN (OOD). All methods achieve similar AUROC ($\sim$0.93 with energy scores), with thicket ensembles performing comparably to the deep ensemble. The energy score consistently outperforms both MSP and mutual information across all ensemble methods.

Notably, the orthogonal thicket achieves MI AUROC of 0.918 compared to 0.894 for the Gaussian variant — a meaningful improvement in the diversity-dependent metric, suggesting that orthogonal

Table 3: OOD detection performance (CIFAR-10 → SVHN). Energy scores outperform mutual information across all methods. Thicket ensembles perform comparably to the deep ensemble. Best results in **bold**; lower FPR@95 is better.

Method	MSP AUROC	Energy AUROC	MI AUROC	Energy FPR@95
Gaussian THICKET	0.913	0.933	0.894	0.389
Orthogonal THICKET	0.914	0.934	**0.918**	0.391
Layer-Scaled THICKET	0.913	0.933	0.901	0.394
DEEP ENSEMBLE	**0.917**	**0.937**	0.908	**0.415**

Table 4: Ensemble size ablation ($\sigma = 0.001$, Gaussian perturbation). Increasing K has no effect because all members are near-identical at this perturbation scale.

K	Clean Acc (%)	PGD-20 Acc (%)
3	94.0	12.5
5	94.1	12.4
10	94.0	12.4

perturbations do create slightly more functional diversity. However, this advantage does not translate to adversarial robustness improvements.

4.4 Ensemble Size Ablation

Table 4 shows that increasing ensemble size K from 3 to 10 has negligible effect on both clean and robust accuracy. Since all members at $\sigma = 0.001$ are near-clones (0.6% disagreement), adding more identical copies cannot improve robustness.

5 Discussion

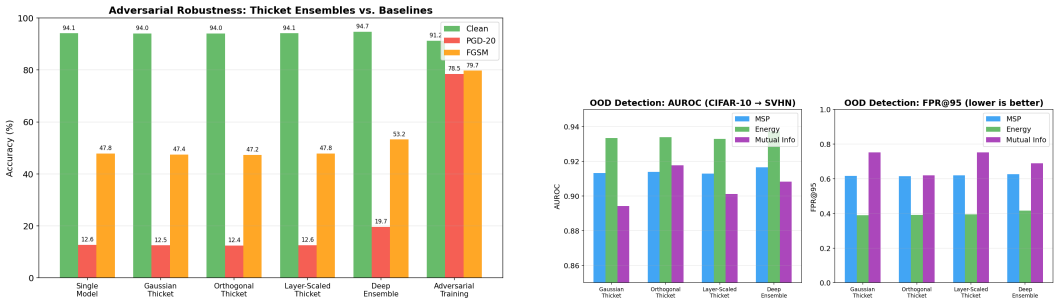
5.1 Why Thicket Ensembles Fail for Vision Robustness

Our results point to a single root cause: RESNET-18 on CIFAR-10 has an extremely tight loss basin. A perturbation of $\sigma = 0.02$ — roughly 2% of parameter magnitudes — is sufficient to destroy the model entirely. This leaves an effective perturbation budget of $\sigma \leq 0.001$, at which scale ensemble members differ by less than 1% of their predictions. A white-box adversary can trivially craft inputs that fool all near-identical members simultaneously.

This finding contrasts with the neural thickets results on LLMs [Gan and Isola, 2026], where models with billions of parameters tolerate substantially larger perturbations. We hypothesize that this difference is driven by overparameterization: LLMs have orders of magnitude more parameters than needed for any individual task, creating broad, flat basins in which many functionally diverse solutions coexist. Vision models like RESNET-18, with ∼11M parameters trained on 50K images, operate much closer to their capacity limit, resulting in sharper basins with less room for diverse nearby solutions.

5.2 The Diversity–Accuracy Tradeoff Is a Cliff, Not a Curve

We expected a smooth tradeoff between accuracy and diversity as σ increases — a regime where moderate diversity could be obtained at modest accuracy cost. Instead, we observe a phase transition: accuracy is essentially preserved up to $\sigma = 0.01$ (92.4%) and then collapses by $\sigma = 0.02$ (28.4%). This cliff-like behavior eliminates the possibility of navigating a Pareto frontier. Neither orthogonal nor layer-scaled perturbation strategies change this picture: the sensitivity is not an artifact of perturbation direction or relative layer scaling, but a fundamental property of the loss landscape geometry.



(a) Adversarial robustness

(b) OOD detection

Figure 2: *(Left)*Adversarial robustness: thicket ensembles match the single model under PGD-20 and FGSM; only adversarial training provides substantial robustness. *(Right)*OOD detection: all ensemble methods achieve similar AUROC with energy scores.

5.3 Deep Ensembles vs. Thicket Ensembles

The deep ensemble (3 independently trained models) achieves PGD accuracy of 19.7% — a 56% relative improvement over the single model. This modest but real robustness gain comes from genuine mode diversity: independently trained models explore different basins of the loss landscape [Fort et al., 2019], leading to meaningfully different decision boundaries. In contrast, thicket ensemble members remain within the same basin, providing no functional diversity for robustness.

However, even the deep ensemble's 19.7% PGD accuracy is far from the 78.5% achieved by adversarial training. This confirms a broader finding in the adversarial robustness literature: ensemble diversity is necessary but far from sufficient for robustness [Pang et al., 2019]. Some form of adversarial awareness during training appears essential.

5.4 OOD Detection: A Qualified Neutral Result

For OOD detection, thicket ensembles perform comparably to deep ensembles (energy AUROC of 0.933 vs. 0.937). This is because energy-based OOD detection primarily depends on the model's confidence calibration rather than ensemble diversity [Xia and Bouganis, 2022]. The average energy score over near-identical members is essentially the single model's energy score. Mutual information — the diversity-dependent metric — performs worse than energy across all methods, consistent with Xia and Bouganis [2022].

The one bright spot is that orthogonal perturbations achieve MI AUROC of 0.918 vs. 0.894 for Gaussian, suggesting that orthogonal perturbations do create marginally more functional diversity even at $\sigma = 0.001$. This is insufficient for adversarial robustness but hints that structured perturbation directions could be valuable in regimes where larger perturbations are tolerable.

5.5 Limitations

Single architecture. We test only RESNET-18 ($\sim$11M parameters). Larger models (e.g., WideResNet-28-10, Vision Transformers) may have broader loss basins that tolerate larger perturbations, potentially enabling effective thicket ensembles.

Single dataset. Our experiments use only CIFAR-10. Higher-resolution datasets (e.g., ImageNet) with correspondingly larger models may exhibit different sensitivity profiles.

White-box attacks only. We evaluate only white-box PGD, where the attacker knows all ensemble members. Transfer attacks between ensemble members might reveal more diversity than the white-box setting captures.

No fine-tuning after perturbation. We test pure perturbation without any gradient-based adaptation. Brief fine-tuning of perturbed models could potentially recover accuracy at larger σ values, enabling meaningful diversity.

Single seed. Due to computational constraints, main experiments use a single seed. The sigma sweep across 60 candidates per scale partially compensates, but formal statistical significance tests require multiple independent runs.

6 Conclusion

We present the first evaluation of neural thicket ensembles for adversarial robustness and OOD detection, testing three perturbation strategies (isotropic Gaussian, orthogonal, layer-scaled) across five scales on RESNET-18/ CIFAR-10. Our central finding is negative but informative: vision models have much tighter loss basins than LLMs, creating a sharp sensitivity cliff that prevents weight-perturbation ensembles from achieving meaningful diversity without catastrophic accuracy loss. At viable perturbation scales, thicket ensemble members are near-clones with $<1\%$ disagreement, providing no adversarial robustness beyond a single model (12.5% vs. 12.6% PGD-20 accuracy). Adversarial training remains vastly superior (78.5%).

The key takeaway for practitioners is clear: do not rely on weight-perturbation ensembles for adversarial robustness in vision models. For the neural thickets research program, our results suggest that the dense thicket phenomenon may be specific to heavily overparameterized models with billions of parameters, where loss basins are broad enough to accommodate functionally diverse solutions within small perturbation neighborhoods.

Several promising directions remain. Testing with larger vision models (ViTs, WideResNets) could reveal whether the sensitivity cliff shifts at scale. Combining thicket perturbation with brief fine-tuning could allow larger perturbations while maintaining accuracy. Perhaps most intriguingly, perturbing adversarially-trained base models — which occupy flatter loss basins [Croce et al., 2023] — may finally unlock the diversity needed for perturbation-based robustness in vision.

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